

Modified 1D Decoupled Near-Field Source Localization: Ambiguity Analysis and Suppression

Yuehe Ning, Jingjing Pan, Jian Chen, Meng Sun, Jianfeng Li, Xiaofei Zhang, and Qixuan Lu

College of Electronic and Information Engineering
Nanjing University of Aeronautics and Astronautics
Nanjing, China
sz2404116@nuaa.edu.cn

Abstract—Near-field source localization requires the joint estimation of direction-of-arrival (DOA) and range. Classical multiple signal classification (MUSIC) relies on an expensive two-dimensional (2D) search. To alleviate this burden, decoupling one-dimensional (1D) search methods have been widely adopted to achieve parameter separation. However, in multi-source scenarios, the decoupled DOA spectrum typically exhibits ambiguity-induced local maxima, especially at low signal-to-noise ratio (SNR), which complicate peak selection. This paper analyzes the origin of such predictable ambiguities inherent in these decoupling methods and proposes a robust two-stage localization scheme. First, a reduced-dimension MUSIC (RD-MUSIC) spectrum is constructed to extract a set of angle candidates, effectively isolating the DOA estimation. Second, each candidate is rigorously re-evaluated in the range domain using the original near-field steering model and ranked by the peak intensity of its corresponding range spectrum. The final DOA–range estimates are obtained by selecting the top-ranked candidates. Simulations and experimental results demonstrate that the proposed scheme effectively suppresses ambiguity-induced peaks and provides robust localization performance.

Index Terms—Near-field localization, MUSIC, reduced dimension, parameter separation.

I. INTRODUCTION

Source localization is a fundamental problem in array signal processing and has been widely applied in sonar, wireless communications, and radar [1]. For far-field sources, the wavefronts are approximately planar so that only the direction-of-arrival (DOA) is required, whereas for near-field sources the wavefronts are spherical and both DOA and range must be estimated [2], [3]. Over the decades, numerous high-resolution algorithms have been developed for far-field localization, such as maximum likelihood estimation (MLE) [4], multiple signal classification (MUSIC) [5], root-MUSIC [6], and estimation of signal parameters via rotational invariance techniques (ES-PRIT) [7].

However, far-field methods built on the planar-wave assumption cannot be directly applied to near-field scenarios due to the non-planar wavefronts. In recent years, various near-field localization methods have been proposed. The MUSIC algorithm is introduced in [8] for near-field source localization and can achieve high estimation accuracy, but with a computationally intensive two-dimensional (2D) search over both DOA and range. To alleviate the computational burden of 2D estimators, several decoupling one-dimensional (1D)

search methods are proposed for near-field source localization, including the second-order statistics (SOS) method [9], the rank reduction (RARE) method [10], and the reduced-dimension MUSIC (RD-MUSIC) algorithm [11]. These decoupling methods mainly exploit the data structure to reduce complexity, such as the anti-diagonal sequence of the covariance matrix [9], [12], or the split of DOA and range in the steering vector [10], [11], [13], [14]. Nevertheless, these 1D search methods depend on symmetric array configuration and the parameter separation gives rise to ambiguous local maxima, especially under low signal-to-noise ratio (SNR).

This paper proposes a modified 1D near-field source localization scheme to address the ambiguity issues inherent in parameter separation. The ambiguity is first analyzed and shown to generate predictable local maxima due to the decoupled steering structure. To suppress these ambiguities, a two-stage method is proposed, which ranks candidate directions based on the peak magnitudes of their corresponding range spectra. Numerical simulations and experimental results are provided to demonstrate both the occurrence of such ambiguities and the effectiveness of their suppression in 1D search spectra.

II. SYSTEM MODEL

Consider K uncorrelated narrowband sources located in the near-field of a symmetric uniform linear array (ULA) with $N = 2M + 1$ sensors (spacing d , wavelength λ), as shown in Fig. 1.

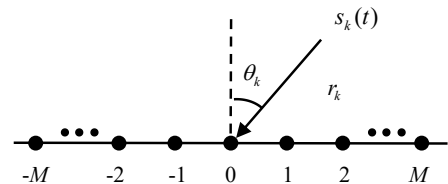


Fig. 1. ULA configuration in near-field source localization.

Taking the array center as the phase reference, the output of the m -th sensor at time t is

$$y_m(t) = \sum_{k=1}^K s_k(t) e^{j\tau_{mk}} + n_m(t), \quad (1)$$

where $s_k(t)$ is the k th source signal; $n_m(t)$ is additive white Gaussian noise, and τ_{mk} is the phase term given by

$$\tau_{mk} \approx m\omega_k + m^2\phi_k, \quad (2)$$

where $\omega_k = -\frac{2\pi d}{\lambda} \sin \theta_k$ and $\phi_k = \frac{\pi d^2}{\lambda r_k} \cos^2 \theta_k$, with θ_k and r_k being the DOA and range of the k -th source, respectively.

In vector form, (1) can be formulated as

$$\mathbf{y}(t) = \mathbf{A}\mathbf{s}(t) + \mathbf{n}(t), \quad (3)$$

where $\mathbf{y}(t)$ is the array output vector; $\mathbf{s}(t)$ is the source signal vector; and $\mathbf{n}(t)$ is the noise vector; $\mathbf{A}$ is the array manifold matrix, with steering vectors $\mathbf{a}(\theta_k, r_k)$ corresponding to each source. Consistent with the phase reference at the array center, the steering vector is given by

$$\mathbf{a}(\theta_k, r_k) = \left[e^{j(-M\omega_k + (-M)^2\phi_k)}, \dots, e^{j(M\omega_k + M^2\phi_k)} \right]^T, \quad (4)$$

where the $(\cdot)^T$ denotes matrix transposition.

The covariance matrix of the received signal estimated with L snapshots is given by

$$\hat{\mathbf{R}} = \frac{1}{L} \sum_{t=1}^L \mathbf{y}(t)\mathbf{y}^H(t), \quad (5)$$

where the $(\cdot)^H$ denotes the conjugate transpose operation. Subsequently, the noise subspace $\mathbf{U}_n \in \mathbb{C}^{N \times (N-K)}$ is obtained via the eigenvalue decomposition (EVD) of $\hat{\mathbf{R}}$.

III. DECOUPLED NEAR-FIELD SIGNAL MODEL

By utilizing the orthogonal relationship between the steering vectors and the noise subspace [8], the 2D-MUSIC spectrum for near-field sources can be written as

$$P_{\text{MUSIC}}(\theta, r) = \frac{1}{\mathbf{a}^H(\theta, r)\mathbf{U}_n\mathbf{U}_n^H\mathbf{a}(\theta, r)}. \quad (6)$$

However, maximizing (6) requires a two-dimensional search over (θ, r) , which is computationally expensive. To alleviate computational complexity, existing reduced-dimension methods, including [10], [11], [13], exploit the inherent symmetry of the array configuration to decouple the steering vector into separate DOA and range components as follows:

$$\begin{aligned} \mathbf{a}(\omega, \phi) &= \underbrace{\begin{bmatrix} e^{j(-M)\omega} & & & \\ & e^{j(-M+1)\omega} & & \\ & & \ddots & \\ & & & 1 \\ & e^{j(M-1)\omega} & & \\ e^{jM\omega} & & & \end{bmatrix}}_{\mathbf{\Gamma}(\omega)} \underbrace{\begin{bmatrix} e^{j(-M)^2\phi} \\ e^{j(-M+1)^2\phi} \\ \vdots \\ e^{j(-1)^2\phi} \\ 1 \end{bmatrix}}_{\mathbf{b}(\phi)} \\ &= \mathbf{\Gamma}(\omega) \mathbf{b}(\phi). \end{aligned} \quad (7)$$

Therefore, $\mathbf{\Gamma}(\omega) \in \mathbb{C}^{(2M+1) \times (M+1)}$ depends only on the DOA parameter, whereas $\mathbf{b}(\phi) \in \mathbb{C}^{(M+1) \times 1}$ contains the range-dependent quadratic phase terms.

Taking RD-MUSIC [11] as an example, the MUSIC denominator is

$$\begin{aligned} J(\theta, r) &= \mathbf{a}^H(\theta, r)\mathbf{U}_n\mathbf{U}_n^H\mathbf{a}(\theta, r) \\ &= \mathbf{b}^H(\phi) \underbrace{\mathbf{\Gamma}^H(\omega)\mathbf{U}_n\mathbf{U}_n^H\mathbf{\Gamma}(\omega)}_{\mathbf{G}(\omega)} \mathbf{b}(\phi), \end{aligned} \quad (8)$$

where $\mathbf{G}(\omega) = \mathbf{\Gamma}^H(\omega)\mathbf{U}_n\mathbf{U}_n^H\mathbf{\Gamma}(\omega)$ is Hermitian positive semidefinite. Here, ω is estimated via the rank-deficiency of $\mathbf{G}(\omega)$ [10] or by minimizing a constrained quadratic form [11]. Subsequently, the optimal ϕ is inferred from the eigenvector corresponding to the smallest eigenvalue of $\mathbf{G}(\omega)$, entirely bypassing the expensive 2D search over (θ, r) .

IV. AMBIGUITY PROBLEM AND PROPOSED METHOD

However, the decoupled DOA spectrum typically exhibits ambiguity-induced local maxima in multi-source scenarios due to the loss of joint information. To analyze this phenomenon, assume there are two sources in the near-field of the array as ω_1 and ω_2 , and $\omega_2 > \omega_1$. Let $\omega_{\text{diff}} = (\omega_2 - \omega_1)/2$ and $\omega_{\text{sum}} = (\omega_1 + \omega_2)/2$ denote the difference and sum of ω , respectively.

Then $\mathbf{\Gamma}(\omega_1)$ can be factored as

$$\mathbf{\Gamma}(\omega_1) = \underbrace{\begin{bmatrix} e^{-j(-M)\omega_{\text{diff}}} & & \\ & \ddots & \\ & & e^{-jM\omega_{\text{diff}}} \end{bmatrix}}_{\mathbf{\Lambda}^*(\omega_{\text{diff}})} \mathbf{\Gamma}(\omega_{\text{sum}}). \quad (9)$$

where $\mathbf{\Lambda}(\omega_{\text{diff}})$ is a diagonal matrix of phase-shift factors, and $(\cdot)^*$ denotes the complex conjugate operator, reflecting the ULA phase symmetry.

Similarly, $\mathbf{\Gamma}(\omega_2)$ can be factorized as

$$\mathbf{\Gamma}(\omega_2) = \mathbf{\Lambda}(\omega_{\text{diff}})\mathbf{\Gamma}(\omega_{\text{sum}}), \quad (10)$$

Substituting these above equations into the reduced-dimension orthogonality condition (i.e., $\mathbf{U}_n^H\mathbf{a}(\theta, r) = \mathbf{0}$), we obtain

$$\begin{aligned} \mathbf{b}^H(\phi_1)\mathbf{\Gamma}^H(\omega_{\text{sum}})\mathbf{\Lambda}(\omega_{\text{diff}})\mathbf{U}_n &= \mathbf{0}, \\ \mathbf{b}^H(\phi_2)\mathbf{\Gamma}^H(\omega_{\text{sum}})\mathbf{\Lambda}^*(\omega_{\text{diff}})\mathbf{U}_n &= \mathbf{0}. \end{aligned} \quad (11)$$

Consequently, there exist matrices $\mathbf{Q}_1$ and $\mathbf{Q}_2$ satisfying

$$\begin{aligned} \mathbf{\Lambda}(\omega_{\text{diff}})\mathbf{U}_n &= \mathbf{U}_n\mathbf{Q}_1, \\ \mathbf{\Lambda}^*(\omega_{\text{diff}})\mathbf{U}_n &= \mathbf{U}_n\mathbf{Q}_2, \end{aligned} \quad (12)$$

where $\mathbf{Q}_1$ and $\mathbf{Q}_2$ are guaranteed to be of full rank. The detailed proof regarding their existence and rank properties is provided in the Appendix.

Using (12) and the fact that $\mathbf{Q}_1, \mathbf{Q}_2$ are nonsingular, (11) reduces to

$$\begin{aligned} \mathbf{b}^H(\phi_1)\mathbf{\Gamma}^H(\omega_{\text{sum}})\mathbf{U}_n &= \mathbf{0}, \\ \mathbf{b}^H(\phi_2)\mathbf{\Gamma}^H(\omega_{\text{sum}})\mathbf{U}_n &= \mathbf{0}. \end{aligned} \quad (13)$$

Thus, ω_{sum} corresponds to a spurious solution satisfying the orthogonality condition, yielding an ambiguity-induced local maximum at

$$\sin(\theta_{\text{local}}) = \frac{\sin(\theta_1) + \sin(\theta_2)}{2}. \quad (14)$$

For $K > 2$, pairwise couplings occur; overlapped ambiguities reduce to distinct peaks, giving at most $K(K+1)/2$ peaks in the spectrum, including at most $\binom{K}{2}$ ambiguity-induced peaks. Notably, this ambiguity is inherent to all 1D methods utilizing the $\mathbf{a}(\omega, \phi)$ decoupling, not limited to RD-MUSIC.

To resolve this structural ambiguity, we propose a two-stage scheme. First, a 1D spatial spectrum is constructed to extract DOA candidates, which may include the aforementioned additional local maxima. Second, each candidate is re-evaluated in the range domain using the near-field quadratic phase term, and the true sources are selected according to a range-consistency criterion.

Stage 1 (DOA Candidate Extraction): Based on the representative RD-MUSIC framework, the decoupled criterion can be formulated as a quadratic optimization problem [11]. Imposing the linear constraint $\mathbf{e}_1^H \mathbf{b}(\phi) = 1$, the minimization admits a closed-form solution, yielding the RD-MUSIC spectrum

$$P_{\text{RD-MUSIC}}(\theta) = \mathbf{e}_1^H \mathbf{G}(\omega)^{-1} \mathbf{e}_1, \quad (15)$$

where $\mathbf{e}_1 = [0, \dots, 0, 1]^T \in \mathbb{C}^{(M+1) \times 1}$ is a selection vector.

By locating the prominent peaks of $P_{\text{RD-MUSIC}}(\theta)$, we obtain a DOA candidate set $\{\tilde{\theta}_\ell\}_{\ell=1}^{L_c}$, where L_c is chosen larger than K to include possible ambiguity-induced local maxima, with $L_c \leq K(K+1)/2$.

Stage 2 (Range-domain Disambiguation): Ambiguity-induced local maxima in the DOA spectrum typically yield weaker responses when validated against the full array manifold. This motivates a range-domain scoring strategy to rank the candidates.

For each DOA candidate $\tilde{\theta}_\ell$ from Stage 1, we construct a 1D spatial spectrum search over the range domain:

$$P_{1D}(\tilde{\theta}_\ell, r) = \frac{1}{\mathbf{a}^H(\tilde{\theta}_\ell, r) \mathbf{U}_n \mathbf{U}_n^H \mathbf{a}(\tilde{\theta}_\ell, r)}, \quad (16)$$

where the search space is restricted to the Fresnel region [15]. The steering vector $\mathbf{a}(\tilde{\theta}_\ell, r)$ is reconstructed using the estimated angle and the scanning range.

The range estimate corresponding to the ℓ -th candidate is obtained by maximizing this spectrum:

$$\tilde{r}_\ell = \arg \max_r P_{1D}(\tilde{\theta}_\ell, r). \quad (17)$$

To distinguish true sources from ambiguity-induced peaks, we define a validity score γ_ℓ as:

$$\gamma_\ell = P_{1D}(\tilde{\theta}_\ell, \tilde{r}_\ell). \quad (18)$$

Finally, by sorting the scores $\{\gamma_\ell\}_{\ell=1}^{L_c}$ in descending order, the indices of the K largest values are selected. The final estimates $\{(\hat{\theta}_k, \hat{r}_k)\}_{k=1}^K$ correspond to these top-ranked candidates.

V. SIMULATIONS AND ANALYSIS

Consider a symmetric ULA with $N = 11$ sensors and inter-element spacing $d = \lambda/4$. Given the array aperture $D = (N-1)d$, the Fresnel region $[0.62\sqrt{D^3/\lambda}, 2D^2/\lambda]$ [15] corresponds to a valid search range of $r \in [2.45\lambda, 12.50\lambda]$.

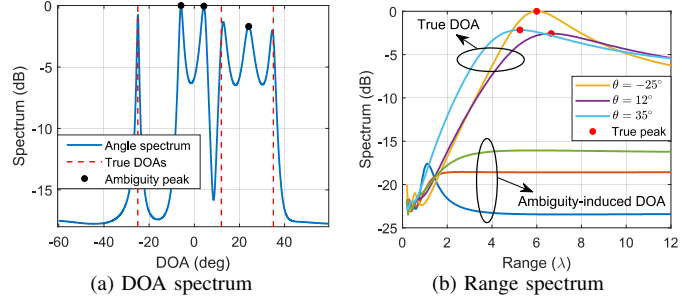


Fig. 2. Simulated DOA and Range Spectra.

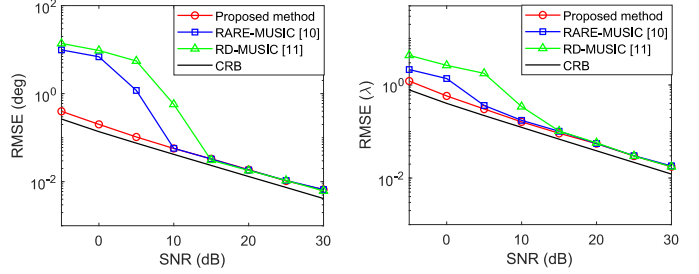


Fig. 3. RMSE performance versus SNR.

Example 1: Fig. 2 gives the results of the proposed algorithm, where $L = 500$ and $\text{SNR} = 0$ dB. Three near-field sources are located at $[-25^\circ, 5.8\lambda]$, $[12^\circ, 7.8\lambda]$, and $[35^\circ, 6.0\lambda]$. As shown in Fig. 2(a), the DOA spectrum captures all true DOAs, while additional ambiguity-induced local maxima arise from the parameter-separation ambiguity predicted by (14). Fig. 2(b) shows the range spectra evaluated at the extracted DOA candidates. Pronounced range peaks are observed for the candidates corresponding to the true DOAs, whereas the ambiguity-induced candidates yield much weaker responses within the considered near-field region. The average running time of the proposed method under the same settings is 9.34s over 100 trials.

Example 2: We evaluate the root mean square error (RMSE) versus SNR (-5 to 30 dB) for two sources at $[-25^\circ, 5.8\lambda]$ and $[32^\circ, 7.8\lambda]$ with $L = 500$. The proposed method is also compared with RARE-MUSIC [10], RD-MUSIC [11], and the Cramér-Rao Bound (CRB) [16] as a theoretical benchmark.

As illustrated in Fig. 3, the proposed method achieves lower RMSE than RARE-MUSIC and RD-MUSIC under low SNR conditions, while maintaining near-CRB accuracy across the whole SNR range. While the benchmark 1D methods degrade significantly due to unresolved ambiguities, our algorithm effectively eliminates false peaks via range-domain disambiguation, maintaining an RMSE close to the CRB across all SNR levels.

VI. EXPERIMENTAL VERIFICATION

To further validate the proposed scheme with real data, we conduct outdoor measurements using a Zynq-based FPGA platform, with the experimental setup similar to that in [17]. As depicted in Fig. 4, the experimental system comprises

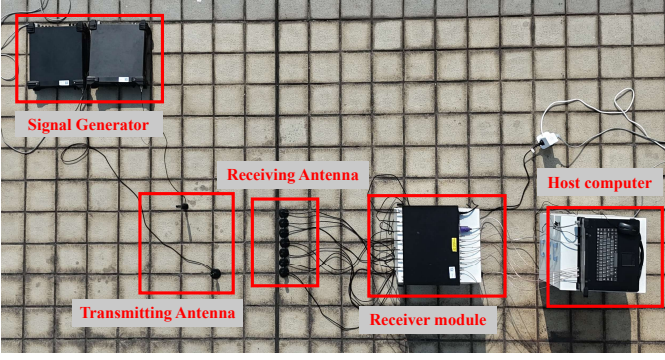


Fig. 4. Experimental setup for near-field localization using a Zynq-based FPGA platform.

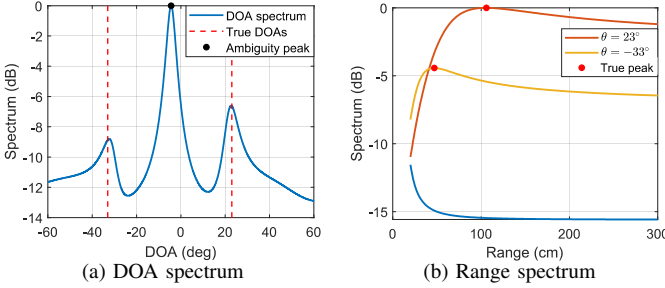


Fig. 5. Measured DOA and Range Spectra.

signal generators, transmitting and receiving antennas, a receiver module, and a host computer for data acquisition and processing. The transmitting antennas are positioned within the Fresnel region, and their DOA and range references are measured using a laser rangefinder and protractor.

The transmitters emit signals at 1.2 GHz and 1.19 GHz (approximate $\lambda \approx 25$ cm). The receiving array is a symmetric ULA with $N = 7$ elements and inter-element spacing $d = 6.25$ cm, defining a Fresnel region of $r \in [28.5, 112.5]$ cm, within which the sources are positioned and their ground-truth parameters measured via a laser rangefinder and protractor. The received signals are downconverted to baseband and synchronously sampled at 100 MHz with 16-bit resolution, collecting $L = 1024$ snapshots per trial. To address channel inconsistencies, an active calibration mechanism is performed to compensate for amplitude and phase errors.

The measured DOA and range spectra are shown in Fig. 5. The true DOA and range values are $[-33^\circ, 23^\circ]$ and $[55\text{cm}, 100\text{cm}]$, respectively, while the estimated DOA values are $[-32.32^\circ, 22.68^\circ]$, with range estimates of $[47\text{cm}, 106\text{cm}]$. Fig. 5(a) shows a local maximum at -4.35° , close to the theoretical local maximum at -4.41° predicted by (14). Similarly, Fig. 5(b) shows prominent peaks in the range spectra corresponding to the true DOAs, with ambiguity-induced local maxima near the theoretical value. These maxima are effectively suppressed by the proposed method. Statistical evaluations over multiple trials yield consistent results, confirming the method's robustness in practical scenarios.

VII. CONCLUSION

In this paper, we propose a modified 1D near-field source localization scheme to resolve the ambiguity problem introduced by parameter separation. By theoretically characterizing the predictable locations of local maxima, a two-stage ambiguity suppression framework is developed. The first stage constructs an RD-MUSIC spectrum to extract DOA candidates, while the second stage re-evaluates these candidates in the range domain using the near-field quadratic phase term. Simulations and experimental results confirm that the method effectively suppresses such ambiguities and maintains robust performance.

APPENDIX

PROOF OF EXISTENCE AND RANK PROPERTY

The validity of the invariance relation in (12) relies on the existence and full-rank property of the coefficient matrix $\mathbf{Q}_1$. We detail the proof for $\mathbf{Q}_1$ below, while the proof for $\mathbf{Q}_2$ is similar.

1) *Proof of Existence:* The noise subspace $\mathbf{U}_n$ is orthogonal to the steering vectors of the true sources, i.e., $\mathbf{a}^H(\theta_k, r_k)\mathbf{U}_n = \mathbf{0}$ for $k = 1, 2$. By substituting the factorization $\mathbf{a}(\theta_1, r_1) = \mathbf{\Lambda}^*(\omega_{\text{diff}})\mathbf{\Gamma}(\omega_{\text{sum}})\mathbf{b}(\phi_1)$ into this condition, and letting $\mathbf{p}_1^H = \mathbf{b}^H(\phi_1)\mathbf{\Gamma}^H(\omega_{\text{sum}})$ denote a vector in the signal subspace $\mathbf{U}_s$, the orthogonality condition can be rewritten as:

$$\mathbf{p}_1^H \mathbf{\Lambda}(\omega_{\text{diff}}) \mathbf{U}_n = \mathbf{0}. \quad (19)$$

Since $\mathbf{\Lambda}(\omega_{\text{diff}})$ is a unitary diagonal matrix, it defines a unitary transformation that preserves the inner product and, consequently, the orthogonality between subspaces. Given the initial condition $\mathbf{U}_s \perp \mathbf{U}_n$, this orthogonality is maintained under the transformation, i.e.,

$$\mathbf{\Lambda}(\omega_{\text{diff}}) \mathbf{U}_s \perp \mathbf{\Lambda}(\omega_{\text{diff}}) \mathbf{U}_n. \quad (20)$$

As $\mathbf{p}_1^H$ belongs to the signal subspace $\mathbf{U}_s$, the fact that its product with $\mathbf{\Lambda}(\omega_{\text{diff}}) \mathbf{U}_n$ vanishes indicates that the columns of the transformed matrix remain within the noise subspace. Furthermore, as the columns of $\mathbf{U}_n$ constitute an orthonormal basis for this subspace, $\mathbf{\Lambda}(\omega_{\text{diff}}) \mathbf{U}_n$ can be uniquely represented as a linear combination of $\mathbf{U}_n$. This fundamentally guarantees the existence of a matrix $\mathbf{Q}_1$ such that

$$\mathbf{\Lambda}(\omega_{\text{diff}}) \mathbf{U}_n = \mathbf{U}_n \mathbf{Q}_1. \quad (21)$$

2) *Proof of Full Rank:* We prove this by contradiction. Suppose $\mathbf{Q}_1$ is singular. Then there exists a non-zero vector $\mathbf{v}$ such that $\mathbf{Q}_1 \mathbf{v} = \mathbf{0}$. Multiplying the invariance equation $\mathbf{\Lambda}(\omega_{\text{diff}}) \mathbf{U}_n = \mathbf{U}_n \mathbf{Q}_1$ by $\mathbf{v}$ yields:

$$\mathbf{\Lambda}(\omega_{\text{diff}}) \mathbf{U}_n \mathbf{v} = \mathbf{U}_n \mathbf{Q}_1 \mathbf{v} = \mathbf{0}. \quad (22)$$

Since $\mathbf{\Lambda}(\omega_{\text{diff}})$ is a diagonal matrix with non-zero exponential entries, it is strictly invertible. Left-multiplying by $\mathbf{\Lambda}^{-1}(\omega_{\text{diff}})$ leads to:

$$\mathbf{U}_n \mathbf{v} = \mathbf{0}. \quad (23)$$

However, $\mathbf{U}_n$ has full column rank (orthonormal columns), which implies that $\mathbf{U}_n \mathbf{v} = \mathbf{0}$ if and only if $\mathbf{v} = \mathbf{0}$. This contradicts the assumption that $\mathbf{v}$ is non-zero. Therefore, $\mathbf{Q}_1$ must be full rank.

REFERENCES

- [1] H. Krim and M. Viberg, "Two decades of array signal processing research: the parametric approach," *IEEE Signal Processing Magazine*, vol. 13, no. 4, pp. 67–94, 1996.
- [2] Z. Chen, G. Gokeda, and Y. Yu, *Introduction to Direction-of-Arrival Estimation*, 2010.
- [3] Z. Zheng, M. Fu, W.-Q. Wang, S. Zhang, and Y. Liao, "Localization of Mixed Near-Field and Far-Field Sources Using Symmetric Double-Nested Arrays," *IEEE Transactions on Antennas and Propagation*, vol. 67, no. 11, pp. 7059–7070, 2019.
- [4] P. Stoica and A. Nehorai, "MUSIC, maximum likelihood and Cramer-Rao bound," in *ICASSP-88, International Conference on Acoustics, Speech, and Signal Processing*, 1988, pp. 2296–2299 vol.4.
- [5] R. Schmidt, "Multiple emitter location and signal parameter estimation," *IEEE Transactions on Antennas and Propagation*, vol. 34, no. 3, pp. 276–280, 1986.
- [6] B. Rao and K. Hari, "Performance analysis of Root-MUSIC," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 37, no. 12, pp. 1939–1949, 1989.
- [7] R. Roy and T. Kailath, "ESPRIT-estimation of signal parameters via rotational invariance techniques," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 37, no. 7, pp. 984–995, 1989.
- [8] Y.-D. Huang and M. Barkat, "Near-field multiple source localization by passive sensor array," *IEEE Transactions on Antennas and Propagation*, vol. 39, no. 7, pp. 968–975, 1991.
- [9] E. Grosicki, K. Abed-Meraim, and Y. Hua, "A weighted linear prediction method for near-field source localization," *IEEE Transactions on Signal Processing*, vol. 53, no. 10, pp. 3651–3660, 2005.
- [10] J. Xie, H. Tao, X. Rao, and J. Su, "Passive localization of noncircular sources in the near-field," in *2015 16th International Radar Symposium (IRS)*, 2015, pp. 493–498.
- [11] X. Zhang, W. Chen, W. Zheng, Z. Xia, and Y. Wang, "Localization of Near-Field Sources: A Reduced-Dimension MUSIC Algorithm," *IEEE Communications Letters*, vol. 22, no. 7, pp. 1422–1425, 2018.
- [12] K. Abed-Meraim, Y. Hua, and A. Belouchrani, "Second-order near-field source localization: algorithm and performance analysis," in *Conference Record of The Thirtieth Asilomar Conference on Signals, Systems and Computers*, vol. 1, 1996, pp. 723–727 vol.1.
- [13] J. Pan, P. R. Singh, and S. Men, "A search-free near-field source localization method with exact signal model," *Journal of Systems Engineering and Electronics*, vol. 32, no. 4, pp. 756–763, 2021.
- [14] J. Pan, S. Lv, Y. Ning, M. Sun, and Y. Wang, "Enhanced Near-Field Coherent Source Localization via Alternating Oblique Projection and Polynomial Root-Finding," *IEEE Sensors Journal*, 2026.
- [15] J. He, L. Li, and T. Shu, "Localization of Near-Field Sources for Exact Source-Sensor Spatial Geometry," *IEEE Signal Processing Letters*, vol. 27, pp. 1040–1044, 2020.
- [16] P. Stoica, E. Larsson, and A. Gershman, "The stochastic CRB for array processing: a textbook derivation," *IEEE Signal Processing Letters*, vol. 8, no. 5, pp. 148–150, 2001.
- [17] J. Li, G. Yang, X. Ren, Z. Fu, and S. Qiu, "Modified ESPRIT and Unitary-Transformation Form for Computationally Efficient DOA Estimation on FPGA," *IEEE Transactions on Instrumentation and Measurement*, vol. 74, pp. 1–9, 2025.